Concave m

Aim:- Determination of:

① Focal length (f)

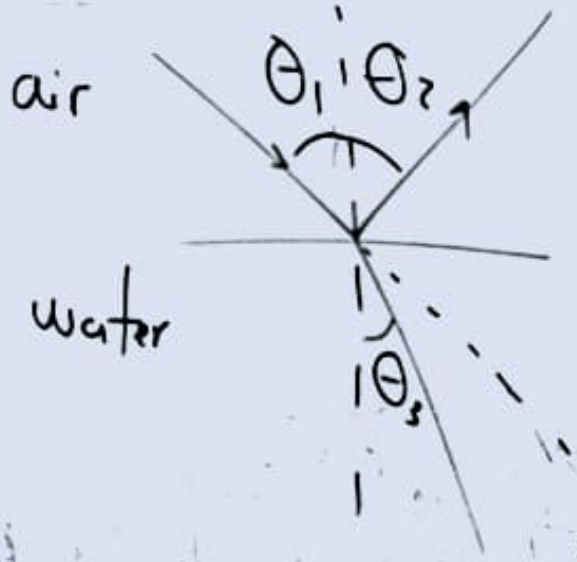
② mirror power (F)

mirrors

planner — مستوية

Spherical — كرية

concave — مقعرة



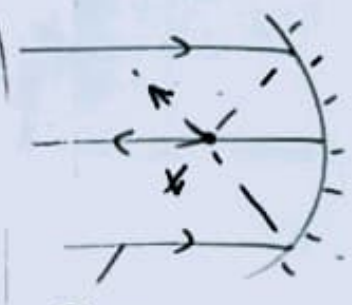
① image Size = object siz

② ~ distance = ~ distan

③ Virtual

④ Inverted

⑤ upright ↑



① Converging

② real

③ downright ↓

④ Infront

⑤ Converging of light

Co

① diverg

② Virtu

③ upright

④ behind

⑤ Conver

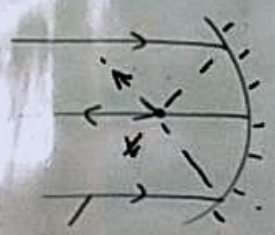
⑥ Reduc

Concave mirror

mirrors

Spherical كروية

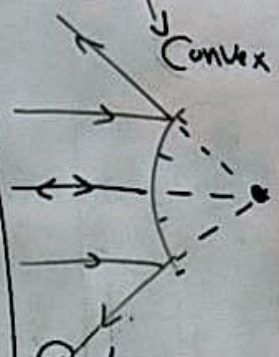
Concave مقعرة



object size ~ distance

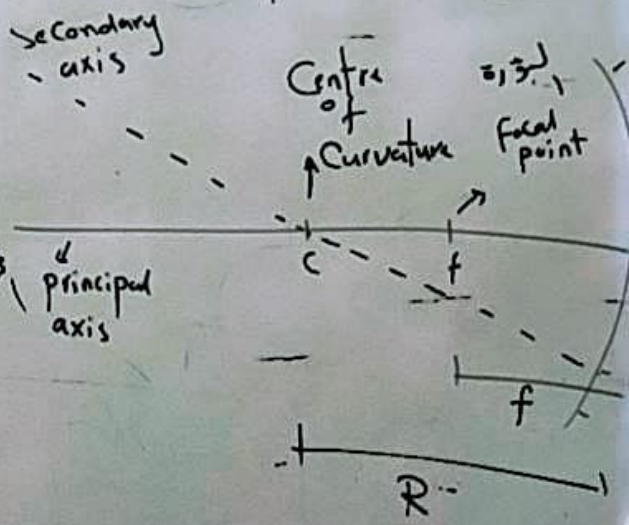
① Converging
② real
③ downright ↓
④ Infront
⑤ Converging of light

Convex محدبة



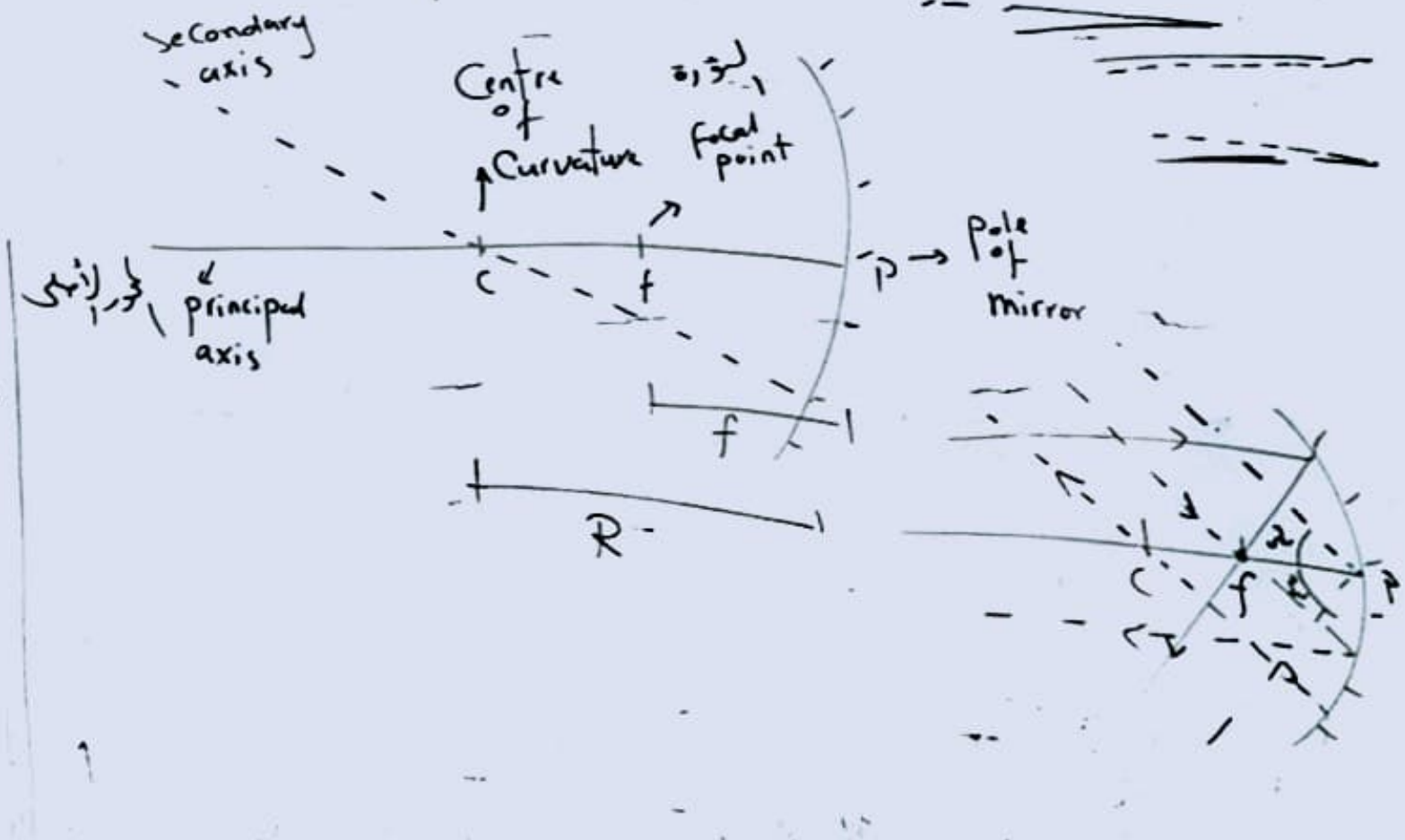
① diverging
② Virtual
③ upright ↑
④ behind
⑤ Converging of the extensions of rays.
⑥ reduced

R=2f



irror

$R = 2f$



nvex

ng

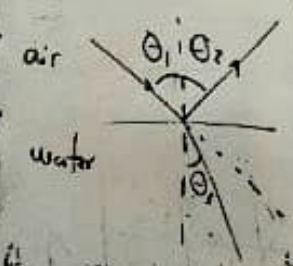
ging of the extensions of rays.

Concave mirror

$R = 2f$

Aim:- Determination of:

1. Focal length (f)
2. mirror power (F)



mirror

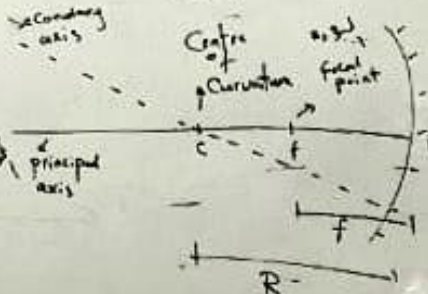
Planar

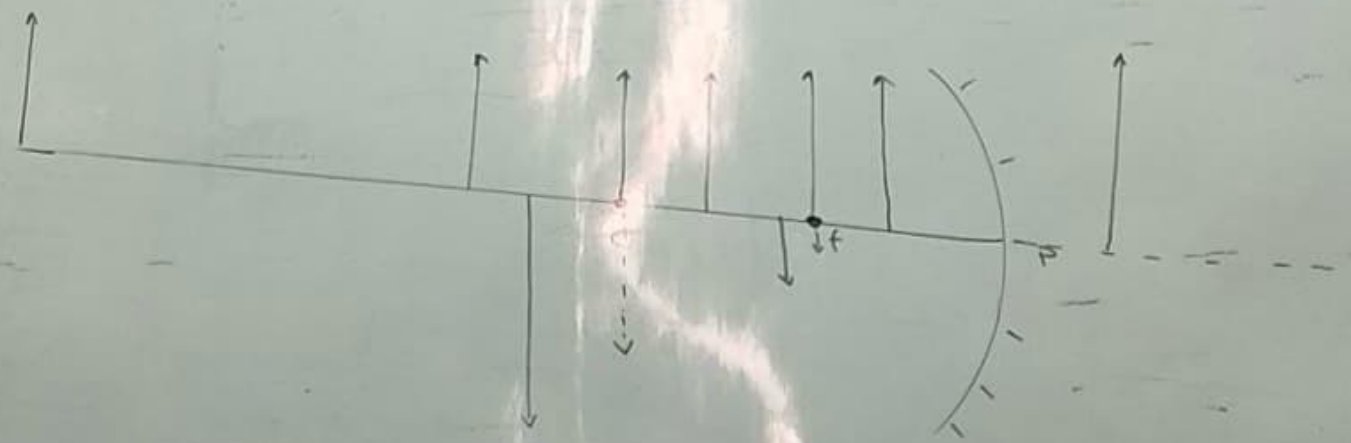
1. image size = object size
2. ~ distance = ~ distance
3. Virtual
4. inverted
5. upright

Spherical

1. Converging
2. real
3. downright
4. inverted
5. Converging of light

1. diverging
2. Virtual
3. upright
4. behind
5. Converging of the extensions of rays.
6. reduced



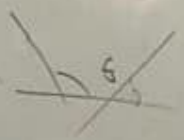

f
P

$f_1 > f_2$ $\quad f_1 = 10\,m$

$F_1 < F_2$ $\quad F_1 = \frac{1}{10\,m} = 0{,}1\,m^{-1} = 0{,}1$ Dioptrie (D)

$f_2 = 10\,cm$

$F_2 = \frac{1}{\frac{10\,cm}{100}} = 10\,m^{-1} = 10\,D$



$$X + Y = F$$

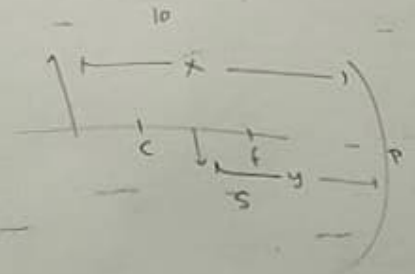
$$Y = -X + F$$

$$\boxed{-5 \pm 5P = 5P}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{f}$$

object distance → x; image → y; focal length → f

$$\frac{1}{10} + \frac{1}{5} = \frac{1}{f}$$



$f_1 > f_2$

$F_1 < F_2$

$f_1 = 10\,m$

$F_1 = \frac{1}{10\,m} = 0.1\,m^{-1} = 0.1$ Diopter (D)

$f_2 = 10\,cm$

$F_2 = \frac{1}{\frac{10\,cm}{100}} = 10\,m^{-1} = 10\,D$

Concave mirror

Aim: ① Determin The Focal length (F)

② " The force of Concave mirror at different distance (F)

Law ① $\frac{1}{F} = \frac{1}{u} + \frac{1}{v}$

② $F = \frac{100}{F}$

Theory ① Types of mirror

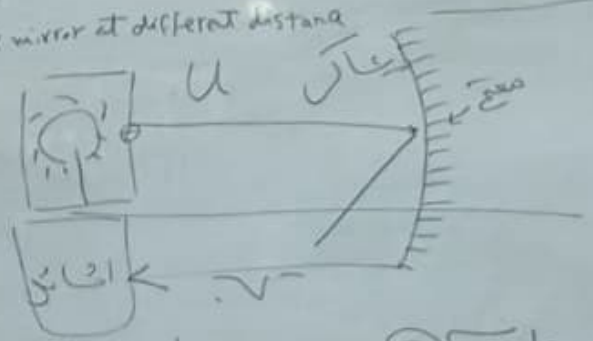
② Focus

③ Focal length

④ Method

Ⓐ Identical Method

Ⓑ General method



$\frac{1}{F} = \frac{1}{u} + \frac{1}{v}$

$u = v$

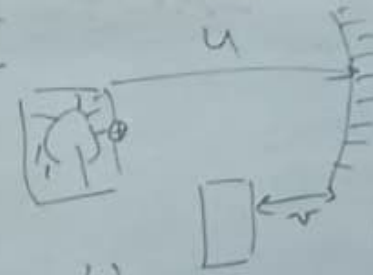
$\frac{1}{F} = \frac{1}{u} + \frac{1}{u}$

$\frac{1}{F} = \frac{2}{u}$

① $F = \frac{u}{2} = \frac{35}{2} = 17.5\,cm$

② $F = \frac{100}{F}$

$F = \frac{100}{17.5} = -cm^{-1}$



$\frac{1}{F} = \frac{1}{u} + \frac{1}{v} \times 100$

$\frac{100}{F} = \frac{100}{u} + \frac{100}{v}$

$F = u + v$

الجهاز			
u	v	$u = \frac{100}{u}$	$v = \frac{100}{v}$
40	15		
45	14		

$u\,(cm^{-1})$

① $F_{avg} = \frac{F_1 + F_2}{2}$

② $F_2 = \frac{100}{F_{avg}}$

F_1

F_2

Focal length (f)

of Concave mirror at different distance

Concave mirror

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$u = v$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{u}$$

$$\frac{1}{f} = \frac{2}{u}$$

① $f = \frac{u}{2} = \frac{35}{2} = 17.5\,cm$

② $F = \frac{100}{f}$

$F = \frac{100}{17.5}$

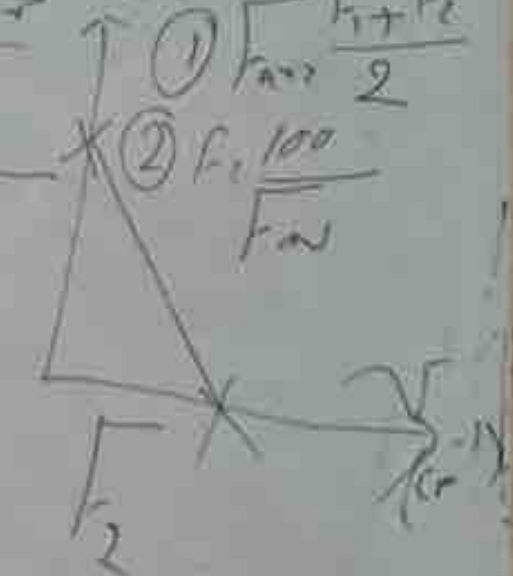
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \times 100$$

$$\frac{100}{f} = \frac{100}{u} + \frac{100}{v}$$

$$F = U + V$$

u	v	$U = \frac{100}{u}$	$V = \frac{100}{v}$
10	15		
15	14		

$U\,(cm^{-1})$

① $F_{avg} = \frac{F_1 + F_2}{2}$

② $F_2 = \frac{100}{F_{avg}}$